Reproducible environment and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Reproducible environment and records

Link exact inputs, script, runtime, metric and raw output without confusing replay with scientific validity.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of reproducible environment and records

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of power and resource tradeoffs

Reference example, not the only acceptable wording: Teacher explanation: Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of predefining an analysis plan

Reference example, not the only acceptable wording: Teacher explanation: An analysis plan records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected.

Deviations remain visible. Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of dataset and benchmark limitations

Reference example, not the only acceptable wording: Teacher explanation: A benchmark measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

Pictured end goal

Explain reproducible environment and records using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Power and resource tradeoffs

Predefining an analysis plan

Dataset and benchmark limitations

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of reproducible environment and records

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient.

Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw **output**s. Verify hashes for integrity without treating them as correctness proofs.

Run-record item	Complete exercise scope
Main folder	Only seed 42 and final-score.txt supplied
Missing main artifacts	Dataset and split, code, preprocessing, dependency/runtime versions, metric/settings and raw outputs
Main conclusion	Replay cannot be established from these materials
Separate mean example	Input 2, 4, 6; sum divided by count
Changed input	2, 4, 9; same formula, new input/run record

Run-record item	Complete exercise scope
Local demonstration	Exact UTF-8 input/script bytes, SHA-256 hashes and actual Python runtime recorded in memory
Randomness	None used in the mean calculation; no seed is needed
Evidence limit	Tiny local replay is not independent AI-model reproduction or proof of valid science

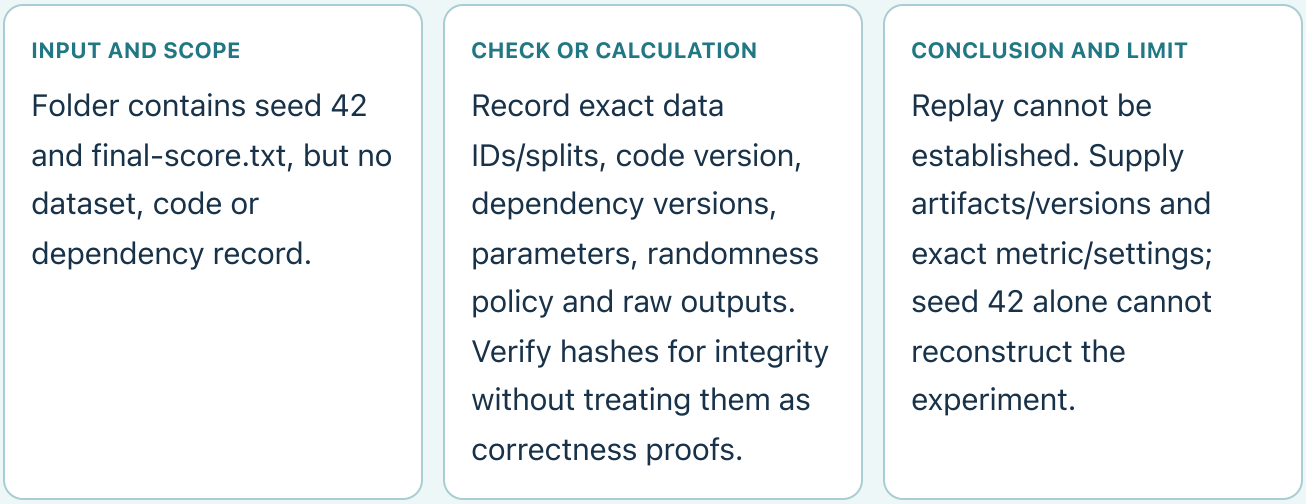
Main evidence ledger and checked key

Main supplied artifact	What it tells us	What remains missing
Seed 42	A stated seed value	Algorithm, input, random library and how seed was used
final-score.txt	A final output record exists in the scenario	Which metric, denominator and run generated it
No dataset/split	Cannot identify evaluated cases	Exact data materials and roles
No code/preprocessing version	Cannot reconstruct transformation	Exact executable procedure
No dependencies/runtime	Cannot establish executable environment	Versions and execution instructions
Replay claim	Not established	Supply artifacts and perform a documented check

Main: seed 42 and final-score.txt do not establish replay; dataset/split, code, preprocessing, versions, metric/configuration and raw outputs are missing. Picture: means are four for 2, 4, 6 and five for 2, 4, 9. The local demonstration executes only that tiny calculation and records actual runtime plus exact-byte input/script hashes. Changed input has a different hash and run ID; the script stays fixed.

Independent case: changed preprocessing and data version require a new run identity, preserved original artifacts/results and a record of the changed versions, settings and lineage. Do not overwrite old evidence or present the new procedure under the old identity. Reproducibility does not establish sound sampling, labels or scientific interpretation.

Reproducible environment and records — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment. Qualification: Reproducibility supports checking the computation, not automatic validity of the research claim. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record.

Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Reproducibility audit	Checked interpretation
Seed and final score only	Missing materials prevent establishing replay
Same seed, different preprocessing	Not the same recorded procedure
Data content changes	Retain original and record changed input/version/run
Hash matches	Checks recorded bytes, not scientific correctness or source history
Same output from two runs	Different inputs can share outputs; output alone does not identify a run
Environment list provided	Still needs materials and execution evidence
Tiny mean re-executed locally	A demonstrated local replay, not an independent full-study reproduction
New run overwrites old ID	Breaks traceability; preserve both versions and lineage

Separate pictured facts and full reasoning

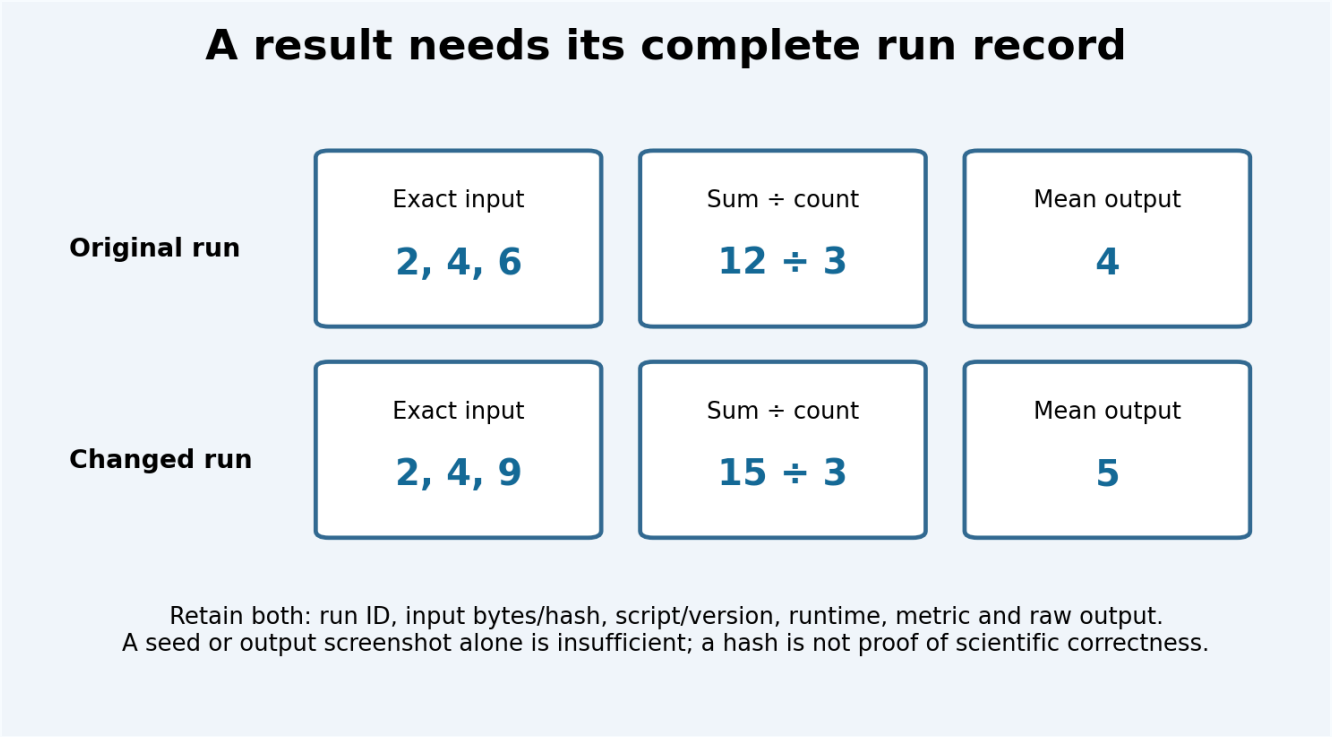
A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.

Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.

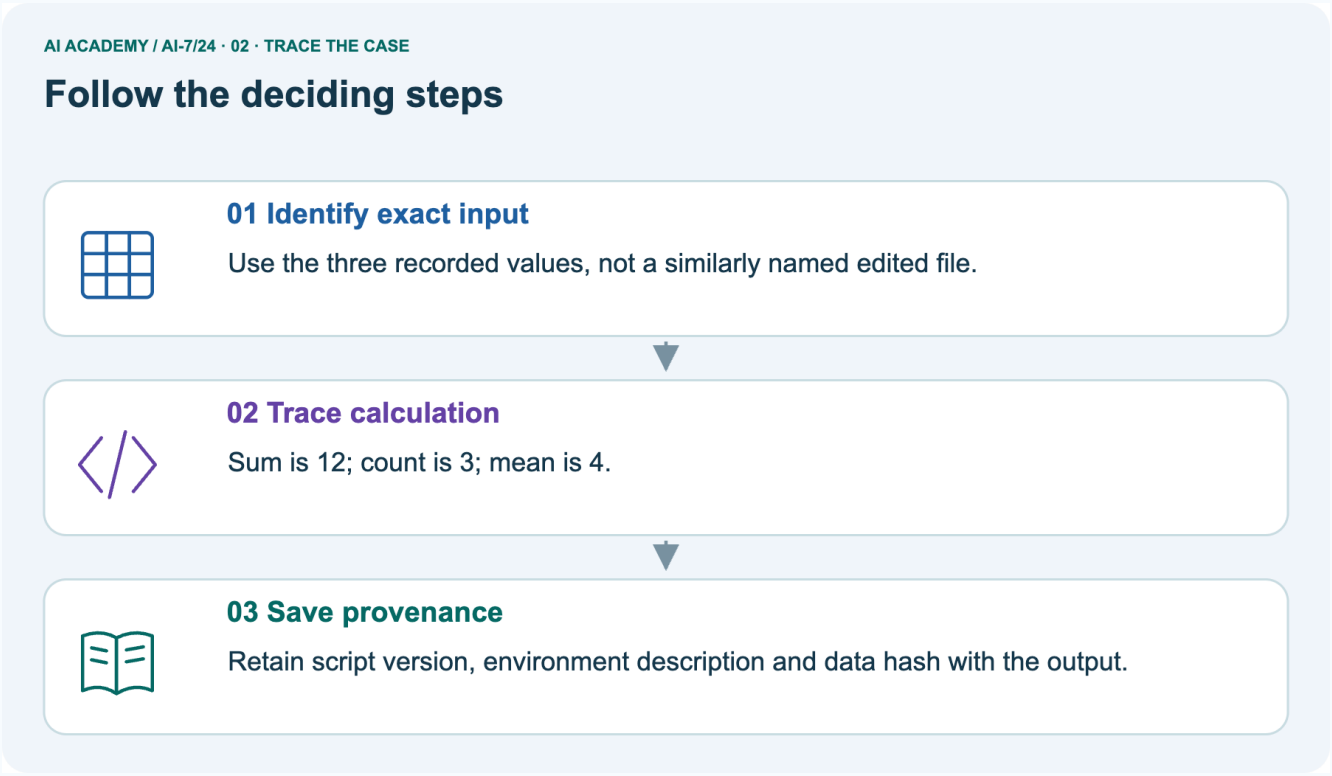
The separate picture records a three-value arithmetic mean. For 2, 4, 6 the sum is twelve and the count is three, so the result is four. Changing the last value to nine gives sum fifteen and mean five. Retain the original record and assign a distinct run identity to the changed input. The accompanying local demonstration actually executes this tiny formula in memory and hashes its exact script and UTF-8 input bytes.

It reports the runtime it is using, not an invented environment. Matching hashes identify those bytes for this audit; they do not prove source history, correct sampling or that the formula answers the research question.

Separate mean record	Original	Changed
Input values	2, 4, 6	2, 4, 9
Sum	12	15
Count	3	3
Exact mean	12/3 = 4	15/3 = 5
Input hash	Hash of original exact bytes	Different bytes produce a different recorded hash here
Script hash	Same recorded formula	Same recorded formula
Run identity	Retained original	New changed-input ID
Interpretation	Local calculation only	Not proof of an AI-model reproduction



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify exact input	Use the three recorded values, not a similarly named edited file.
2. Trace calculation	Sum is 12; count is 3; mean is 4.
3. Save provenance	Retain script version, environment description and data hash with the output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

Change: The data file changes to 2,4,9.

Prediction: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Environment manifest	Runtime and dependency records
Artifact hash	A digest for comparing exact content
Run identity	A versioned link between settings and outputs

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reproducible environment and records

The facts we will use

AI ACADEMY / AI-7/24

Recorded mean calculation

	Value 1	Value 2	Value 3	Mean
Original data	2	4	6	4
Changed data	2	4	9	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/24

Reproducible environment and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify exact input

Use the three recorded values, not a similarly named edited file.



2. Trace calculation

Sum is 12; count is 3; mean is 4.



3. Save provenance

Retain script version, environment description and data hash with the output.

CHANGE ONE THING

The data file changes to 2,4,9.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Reproducible environment and records. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Reproducibility supports checking the computation, not automatic validity of the research claim.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

Use the three recorded values, not a similarly named edited file. Sum is 12; count is 3; mean is 4. Retain script version, environment description and data hash with the **output**.

Changed case: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

Changed-case reasoning: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main: seed 42 and final-score.txt do not establish replay; dataset/split, code, preprocessing, versions, metric/configuration and raw outputs are missing. Picture: means are four for 2, 4, 6 and five for 2, 4, 9. The local demonstration executes only that tiny calculation and records actual runtime plus exact-byte input/script hashes. Changed input has a different hash and run ID; the script stays fixed.

Independent case: changed preprocessing and data version require a new run identity, preserved original artifacts/results and a record of the changed versions, settings and lineage. Do not overwrite old evidence or present the new procedure under the old identity. Reproducibility does not establish sound sampling, labels or scientific interpretation.

Diagnose the current misconceptions

The worked result establishes every future case: Reproducibility supports checking the computation, not automatic validity of the research claim.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

Teacher explanation: Preserve original artifacts/results, assign a new run/version and document changes. Different **input**s define a different comparison.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient.

Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw **output**s. Verify hashes for integrity without treating them as correctness proofs.

W2 Trace the worked case

Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

W3 Audit the limit

A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Reproducibility supports checking the computation, not automatic validity of the research claim.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Reproducibility supports checking the computation, not automatic validity of the research claim.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment. Qualification: Reproducibility supports checking the computation, not automatic validity of the research claim. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

Teacher explanation: Preserve original artifacts/results, assign a new run/version and document changes. Different **input**s define a different comparison.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Preserve original artifacts/results, assign a new run/version and document changes. Different **input**s define a different comparison. Require intermediate reasoning and the stated scope.

Fresh retry: File renamed but bytes unchanged; then one value edited. Expected reasoning: Content hash unaffected by rename, changed by edited bytes; neither hash verifies sound labels or design.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled reproducible environment and records **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reproducible environment and records.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Running the primary experiment. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/25, ai-7/27, ai-7/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of reproducible environment and records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

Check every learner before discussion: Use the worked case for Reproducible environment and records. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Reproducibility supports checking the computation, not automatic validity of the research claim.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The data file changes to 2,4,9. Require a **prediction** and a reason before comparing. Expected result: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

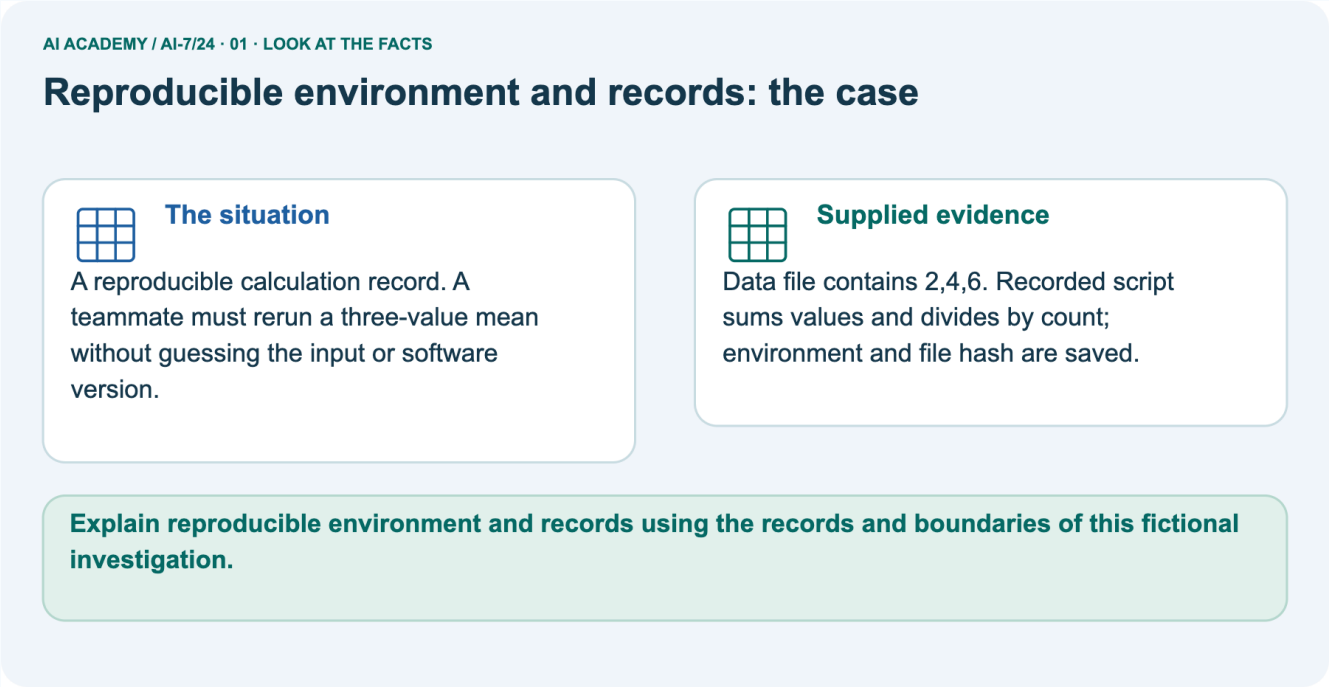
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Reproducible environment and records: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

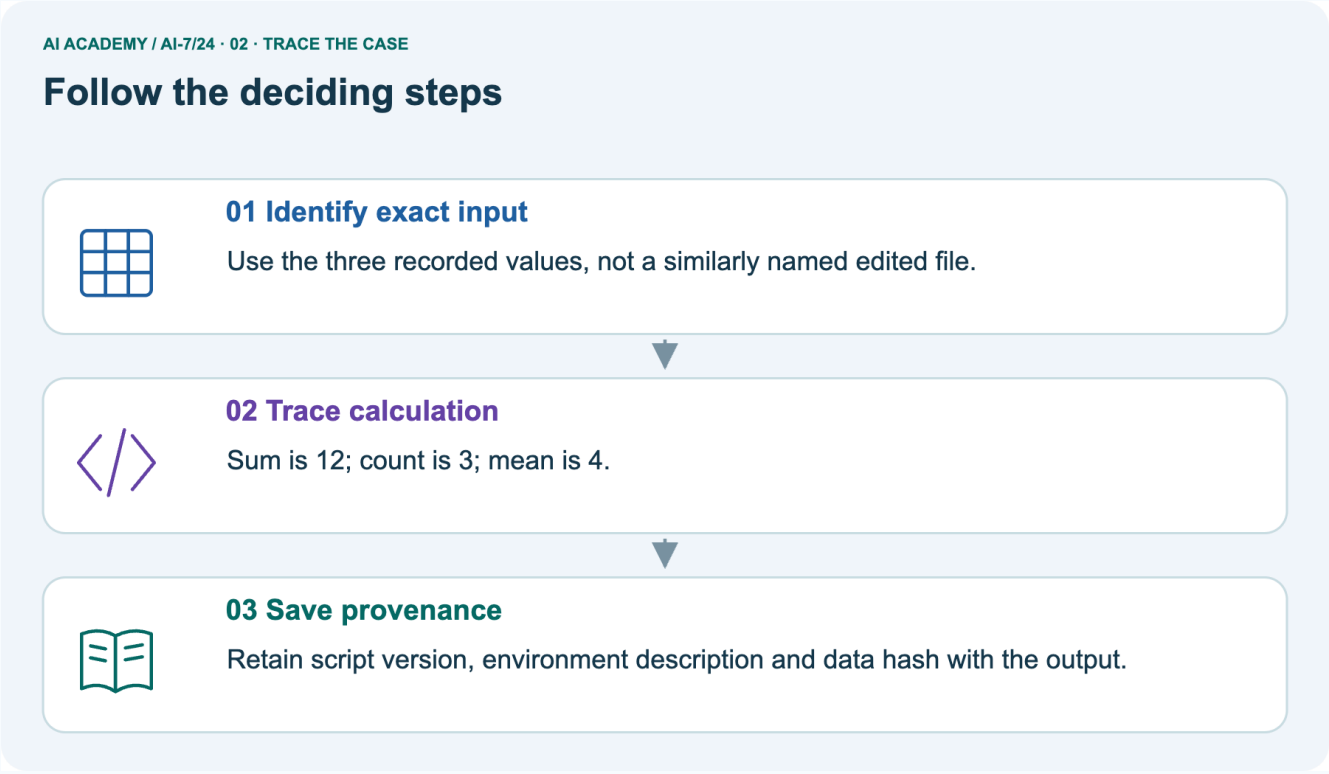
A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version. Supplied facts: Data file contains 2,4,6.

Recorded script sums values and divides by count; environment and file hash are saved.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



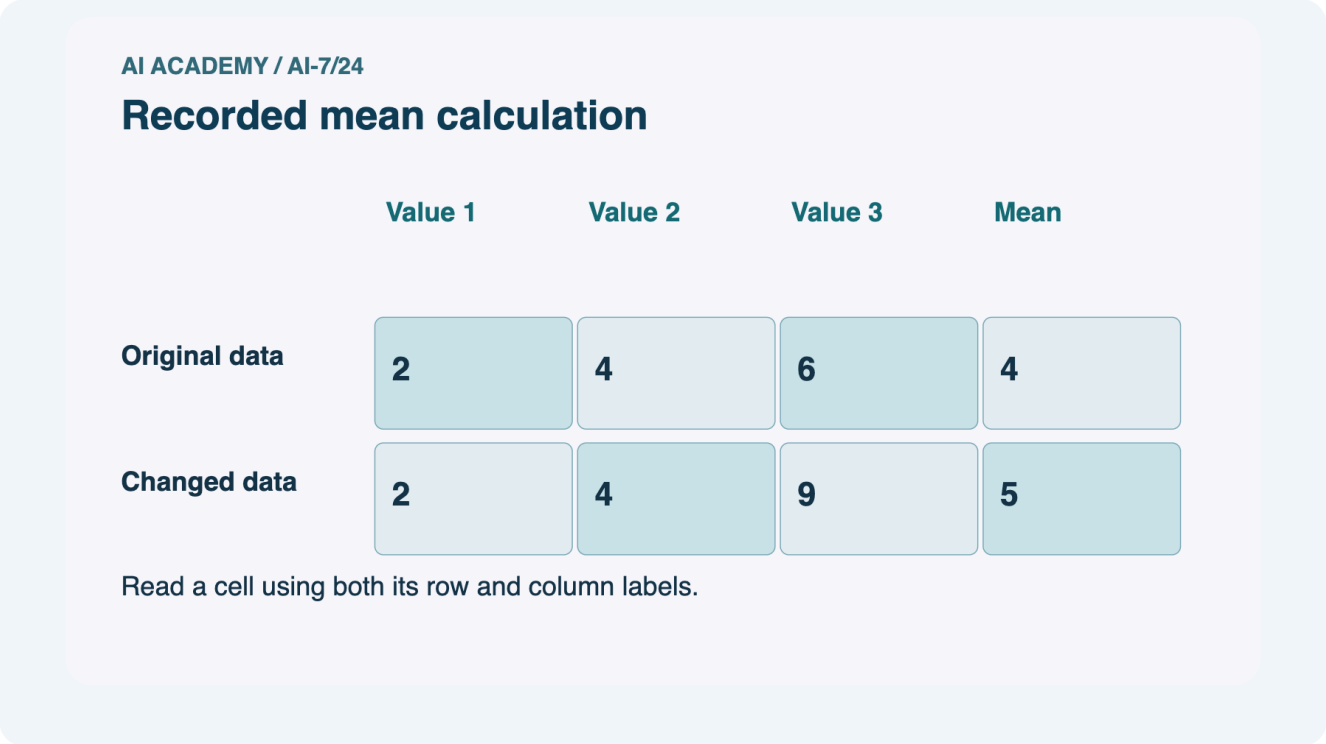
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify exact **input**: Use the three recorded values, not a similarly named edited file. → Trace calculation: Sum is 12; count is 3; mean is 4. → Save provenance: Retain script version, environment description and data hash with the **output**.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Recorded mean calculation



Follow the labelled picture with the matching explanation. Open the vector diagram

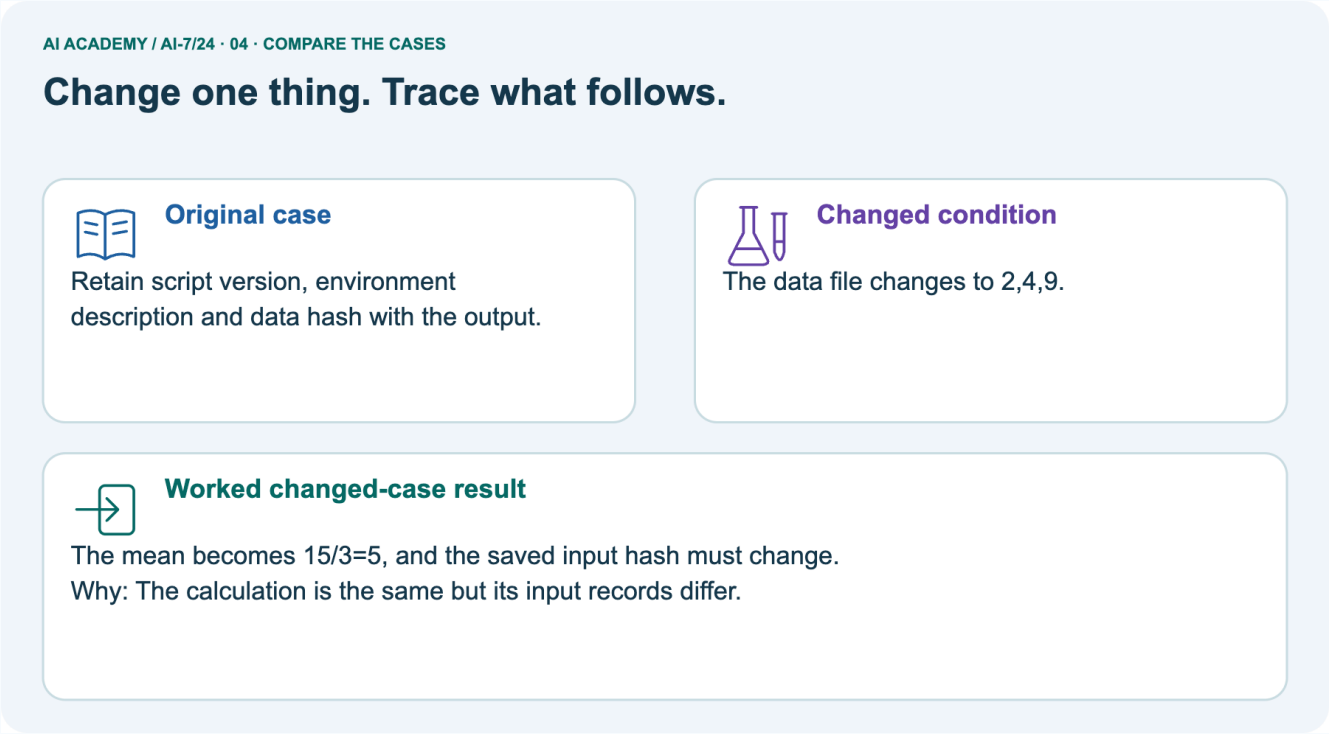
Recorded mean calculation. Original data: Value 1 = 2, Value 2 = 4, Value 3 = 6, Mean = 4. Changed data: Value 1 = 2, Value 2 = 4, Value 3 = 9, Mean = 5.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain script version, environment description and data hash with the **output**. Changed condition: The data file changes to 2,4,9. Expected result: The mean becomes 15/3=5, and the saved **input** hash must change.

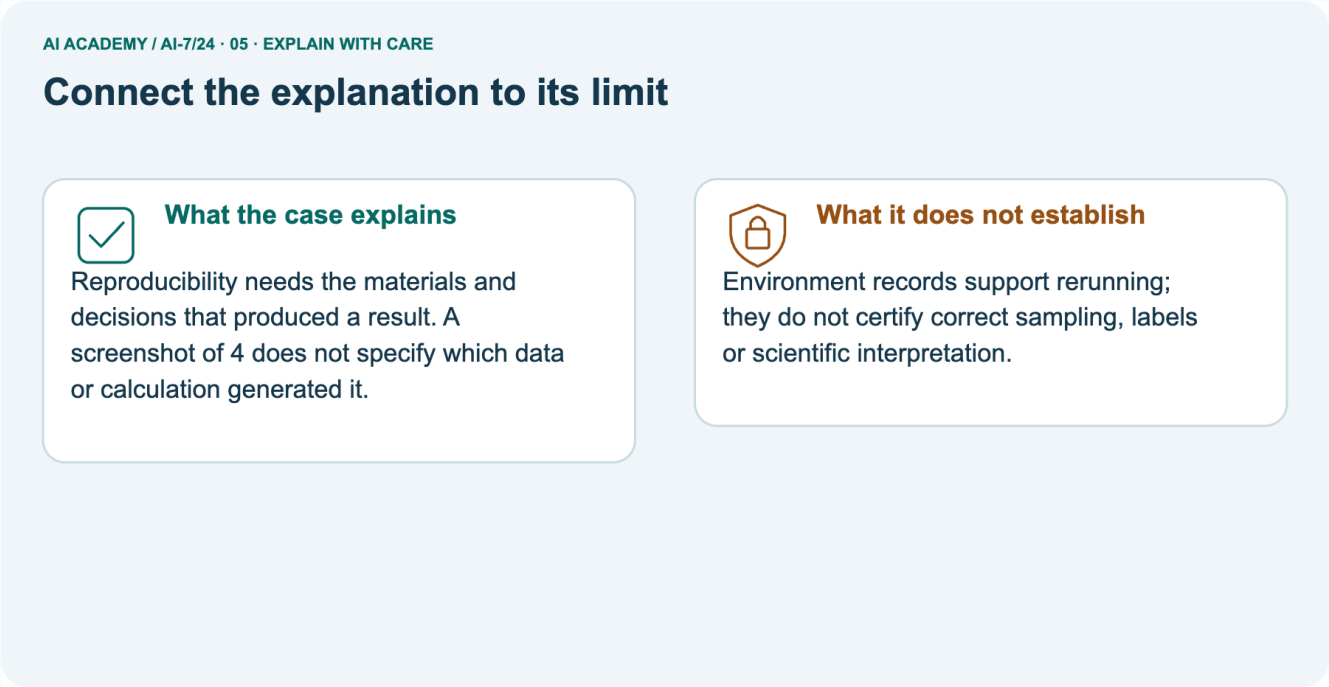
Reason: The calculation is the same but its **input** records differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The calculation is the same but its **input** records differ.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

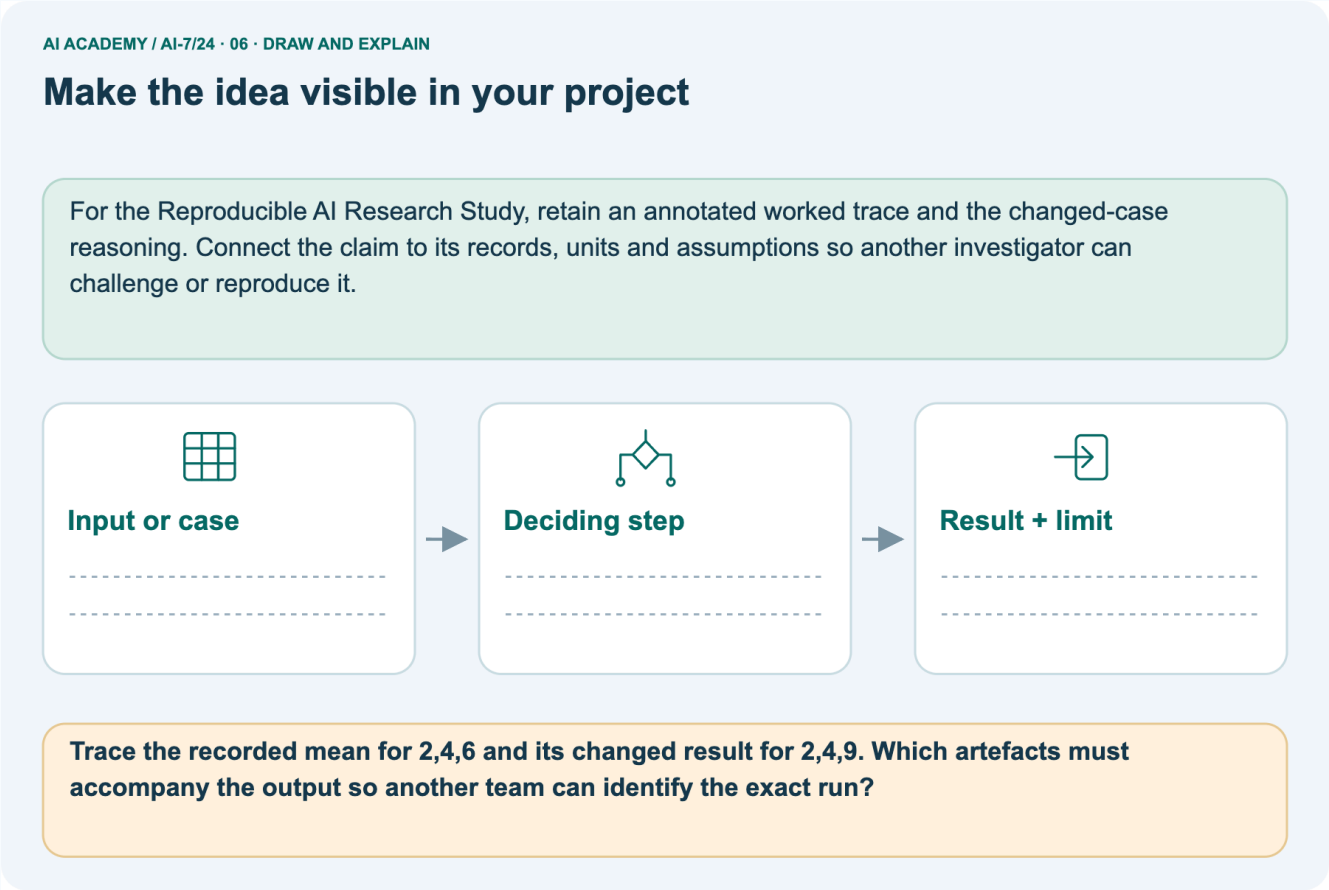
Limit: Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use the three recorded values, not a similarly named edited file. Sum is 12; count is 3; mean is 4. Retain script version, environment description and data hash with the **output**.

Changed case: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Reproducible environment and records: the case

02. Follow the deciding steps

03. Recorded mean calculation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.